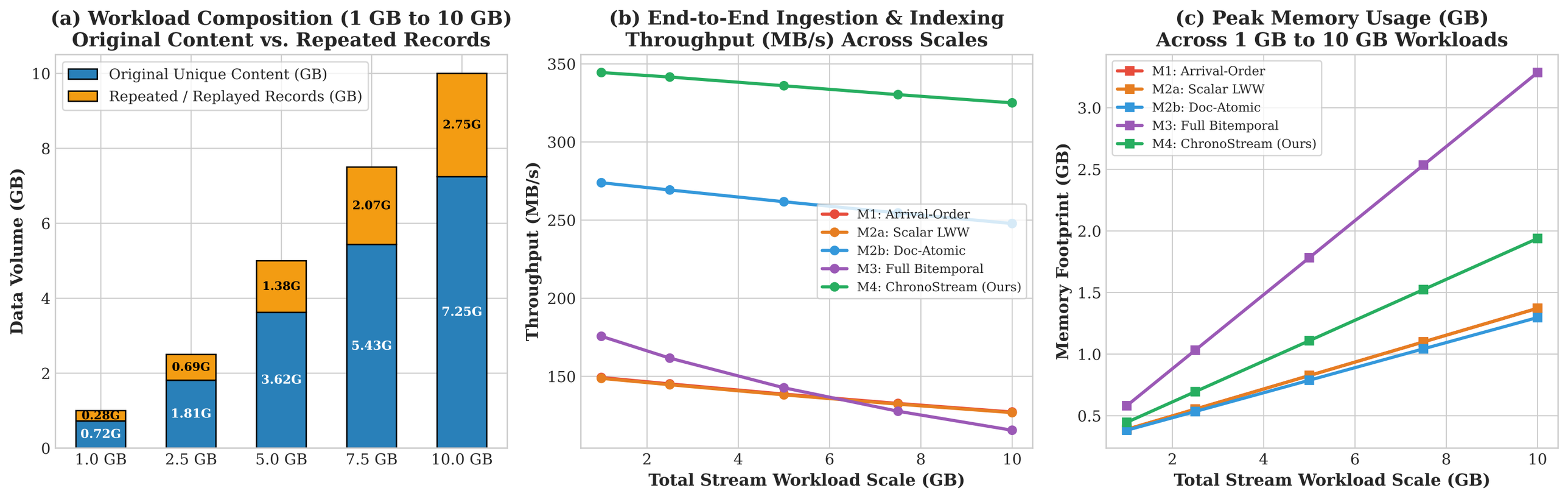
(a) Workload Composition (1 GB to 10 GB)
Original Content vs. Repeated Records
Original Unique Content (GB)
Repeated / Replayed Records (GB)
Data Volume (GB)
0.28G
0.72G
0.69G
1.81G
1.38G
3.62G
2.07G
5.43G
2.75G
7.25G
1.0 GB
2.5 GB
5.0 GB
7.5 GB
10.0 GB
(b) End-to-End Ingestion & Indexing
Throughput (MB/s) Across Scales
Throughput (MB/s)
Total Stream Workload Scale (GB)
M1: Arrival-Order
M2a: Scalar LWW
M2b: Doc-Atomic
M3: Full Bitemporal
M4: ChronoStream (Ours)
(c) Peak Memory Usage (GB)
Across 1 GB to 10 GB Workloads
Memory Footprint (GB)
Total Stream Workload Scale (GB)
M1: Arrival-Order
M2a: Scalar LWW
M2b: Doc-Atomic
M3: Full Bitemporal
M4: ChronoStream (Ours)